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Alterations in gut-derived uremic toxins before the onset of azotemic chronic kidney disease in cats

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Abstract

Background: Although gut-derived uremic toxins are increased in azotemic chronic kidney disease (CKD) in cats and implicated in disease progression, it remains unclear if augmented formation or retention of these toxins is associated with the development of renal azotemia.**Objectives:** Assess the association between gut-derived toxins (ie, indoxyl-sulfate, p-cresyl-sulfate, and trimethylamine-N-oxide [TMAO]) and the onset of azotemic CKD in cats.**Animals:** Forty-eight client-owned cats.**Methods:** Nested case-control study, comparing serum and urine gut-derived uremic toxin abundance at 6-month intervals between initially healthy cats that developed azotemic CKD (n = 22) and a control group (n = 26) that remained healthy, using a targeted metabolomic approach.**Results:** Cats in the CKD group had significantly higher serum indoxyl-sulfate (mean [SD], 1.44 [1.06] vs 0.83 [0.46]; $P = .02$) and TMAO (mean [SD], 1.82 [1.80] vs 1.60 [0.62]; $P = .01$) abundance 6 months before the detection of azotemic CKD. Furthermore, logistic regression analysis indicated that indoxyl-sulfate (odds ratio [OR]: 3.2; 95% confidence interval [CI]: 1.2–9.0; $P = .04$) and TMAO (OR: 3.9; 95% CI: 1.4–11; $P = .03$) were predictors for the onset of azotemia 6 months before diagnosis. However, renal function biomarkers creatinine, symmetric dimethylarginine, and urinary specific gravity were significantly correlated with indoxyl-sulfate and TMAO abundance, causing a loss in predictive significance after correction for these factors.**Conclusions:** Impaired gut-derived uremic toxin handling is apparent at least 6 months before the diagnosis of azotemia, likely reflecting an already ongoing decrease in GFR, tubular function, or both. A direct causal relationship between gut-derived uremic toxicity and the initiation of CKD in cats is still lacking.**Abbreviations:** CKD, chronic kidney disease; IQR, inter quartile range; IRIS, International Renal Interest Society; FDR, false discovery rate; GFR, glomerular filtration rate; QCS, quality Control samples; SDMA, symmetric dimethylarginine; sCr, serum creatinine; TMA, trimethylamine; TMAO, trimethylamine-N-oxide; UHPLC-HRMS, ultrahigh-performance liquid chromatography-high-resolution mass spectrometry; UPC, urine protein to creatinine ratio; USG, urine-specific gravity.This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.© 2024 The Author(s). *Journal of Veterinary Internal Medicine* published by Wiley Periodicals LLC on behalf of American College of Veterinary Internal Medicine.

KEYWORDS

CKD development, early intervention, feline chronic kidney disease, pathophysiology, uremic toxicity

1 | INTRODUCTION

In CKD of cats, it is well-established that circulating concentrations of specific gut-derived uremic toxins (indoxyl-sulfate, p-cresyl-sulfate, and trimethylamine-N-oxide [TMAO]) start to accumulate with declining renal function.^{1,2} The formation of such toxins involves intestinal bacterial modification from dietary precursors.¹⁻⁵ Because these toxins (especially indoxyl-sulfate) are well-recognized mediators of tubular inflammation and have been associated with progression to advanced renal dysfunction,⁶⁻⁸ therapeutic reduction of retained uremic toxins has been an important topic in recent research.^{9,10} Interestingly, different approaches for decreasing uremic toxin concentrations in cats with CKD have been introduced including: decreasing dietary protein content,¹¹ altering the gut microbiome,^{12,13} and binding intestinal toxin precursors.¹⁴

Despite the observation that uremic toxins accumulate in azotemic CKD in cats and are implicated in disease progression, it remains unclear if increased formation or retention of these toxins is associated with the development of renal azotemia. In the progression from initial functional decline to azotemic CKD, a high dietary protein content typical of diets for cats may play a considerable role.^{11,15,16} Therefore, it can be hypothesized that formation or retention of specific gut-derived uremic toxins derived from dietary precursors highly abundant in the diet may underlie this observation in the early disease process.^{17,18} Addressing this knowledge gap may provide supporting argumentation for the potential involvement of these toxins in the initial pathological mechanisms underlying CKD in cats and may support the use of toxin lowering strategies in early disease stages.

Therefore, our main aim was to investigate the relationship between gut-derived uremic toxins and the development of renal azotemia in cats. We focused on measuring serum and urine abundance of indoxyl-sulfate, p-cresyl-sulfate, and TMAO, as well as their dietary precursors, in cats before the onset of azotemic CKD.¹⁹⁻²¹ By conducting longitudinal follow-up of healthy senior cats, we hoped to identify differences in gut-derived uremic toxin pathways between cats that developed azotemic CKD and those that remained healthy. Using a targeted metabolomic approach we specifically intended to determine if increased abundance of these toxins could predict the onset of azotemic CKD. Ultimately, our goal was to establish whether impaired handling of gut-derived uremic toxins is evident before the diagnosis of azotemic CKD in cats.

2 | MATERIALS AND METHODS

2.1 | Ethics statement

The study protocol was approved by the ethical committee of the faculties of Veterinary Medicine and Bioscience Engineering

(EC2020-081). All specimens from cats in the study were obtained prospectively from privately owned domestic cats using a proactive recruitment process. Written informed consent was obtained before inclusion in the study.

2.2 | Study design

Our nested case-control study comprised a prospective cohort investigation involving a target population of healthy domestic senior cats (>10 years of age) at baseline. The evaluated outcome for the target population was development of azotemic CKD. The key criteria for CKD diagnosis were serum creatinine concentration (sCr) >1.83 mg/dL (>161.8 µmol/L), urinary specific gravity (USG) <1.035, and compatible history. The study population consisted of initially healthy senior cats that were recruited via the research facility of the Small Animal Clinic at the Faculty of Veterinary Medicine, between September 2021 and March 2023. Health at baseline was determined by a physical examination, systolic blood pressure (SBP) measurement (Doppler method), CBC, serum biochemistry (including sCr, symmetric dimethylarginine [SDMA], electrolytes), feline leukemia virus antigen testing (FeLV), feline immunodeficiency virus serology (FIV), and complete urinalysis including urinary protein-to-creatinine ratio (UPC) determination and bacterial culture, following established protocols.²²⁻²⁷ The cats were screened every 6 months using the same tests (except FIV and FeLV screening). The study population was further defined based on health reevaluation every 6 months, with development of azotemic CKD as the outcome of interest. Cats showing evidence of any other clinically relevant health abnormalities (except situational hypertension) on the listed examinations at baseline or during follow-up were excluded.

The study group consisted of:

1. A CKD development cohort including all cats from the study population that developed azotemic CKD during the study period based on the defined criteria (sCr >1.83 mg/dL, USG <1.035).²⁸ Data on cats that developed CKD was tracked back to 6 and 12 months before their diagnosis (ie, timepoints –12 months, –6 months, and time of CKD onset). Cats suspected of acute kidney injury were not included. The sCr cut-off in the study was based on a previously established laboratory-specific reference interval consisting of 130 healthy cats.²⁹
2. A control cohort of randomly selected cats from the study population that remained healthy throughout 1 year of follow-up. Data on cats that remained healthy was tracked back to 6 and 12 months before health confirmation (ie, timepoints –12 months, –6 months, and maintained health). Additionally,

cats with renal function abnormalities (sCr >1.83 mg/dL, SDMA >14 µg/dL, USG <1.035, UPC >0.3 or a clinically relevant increase over time in sCr, SDMA, or both [$>25\%$] within the reference interval) were not eligible for inclusion in the control cohort.²⁷ Controls were matched to cases in an approximately 1 : 1 ratio by simple random sampling. The general study design is presented in Figure 1.

All cats, regardless of their cohort, were fed commercially available diets. Specific renal formula diets were introduced only after the onset of azotemic CKD.

The primary objective of our nested case-control study was to assess the association between circulating gut-derived uremic toxins and the development of azotemic CKD before diagnosis. Furthermore, the longitudinal evolution in gut-derived uremic toxin metabolism and

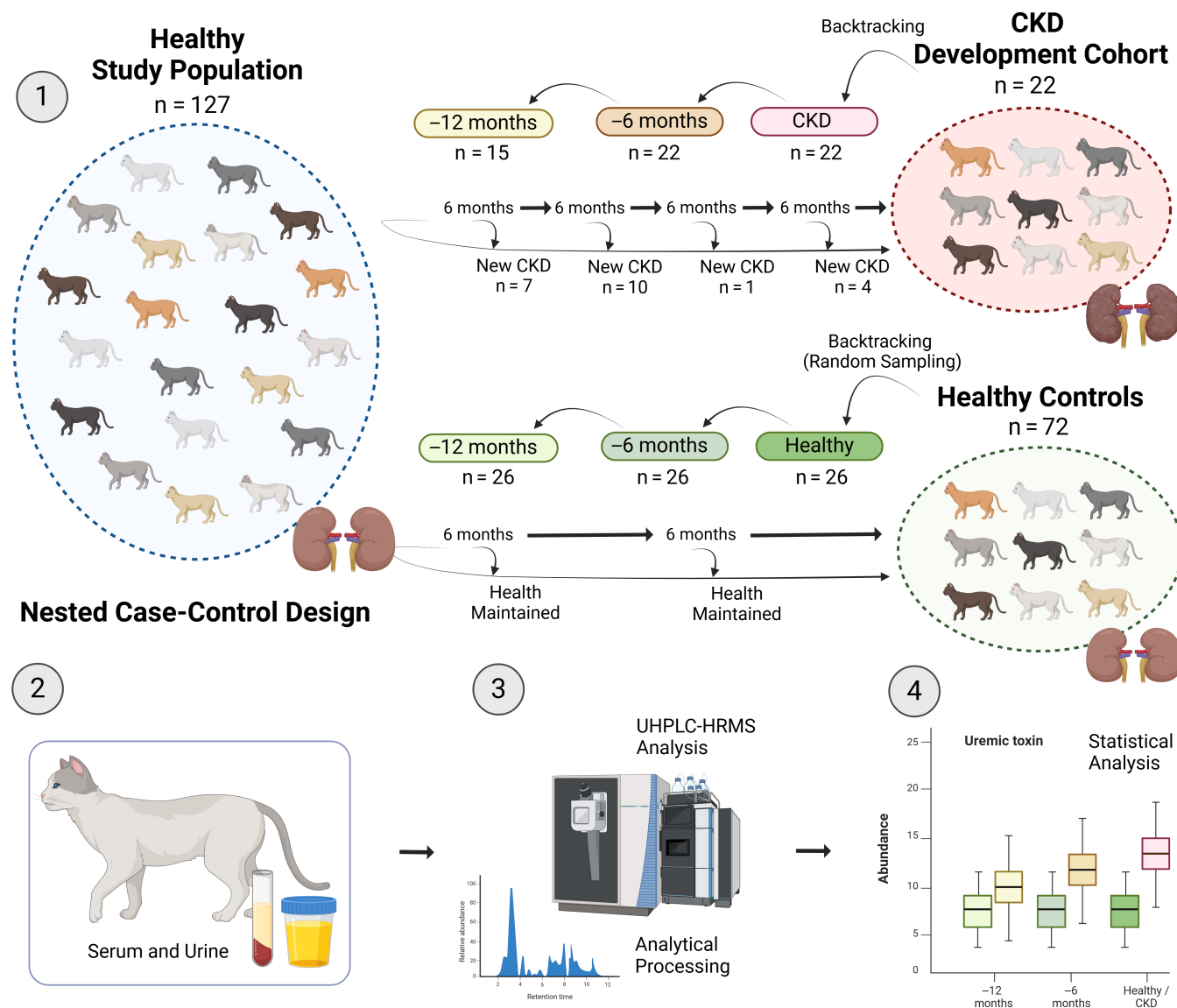


FIGURE 1 (1) This prospective nested case-control study consisted of a healthy population of cats that was followed longitudinally, with assessments conducted at 6-month intervals. Group 1 consisted of all cats that developed azotemic CKD during the study (further referred to as CKD development cohort). Data on cats that developed CKD was tracked back to 6 and 12 months before their diagnosis (ie, timepoints -12 months, -6 months, and CKD onset). Group 2 consisted of a control cohort of cats that remained healthy throughout 1 year follow-up. Likewise, data on cats that remained healthy was tracked back to 6 and 12 months before health confirmation (ie, timepoints -12 months, -6 months, and maintained health). (2) Serum and urine samples were collected, followed by chemical extraction according to feline specific optimized and validated methods. (3) Samples from both matrices were analyzed using ultrahigh-performance liquid chromatography coupled to high-resolution mass spectrometry (UHPLC-HRMS). (4) The primary objective of our nested case-control study was to assess the association between circulating gut-derived uremic toxins and the development of azotemic CKD, 6 and 12 months before diagnosis. Statistical analyses also aimed to compare metabolite abundance between the CKD development cohort and the healthy cohort (at timepoints -12 months, $n = 15$ vs $n = 26$; -6 months, $n = 22$ vs $n = 26$ and CKD onset or maintained health, $n = 22$ vs $n = 26$). Furthermore, the trend of altering metabolites was evaluated over time in the CKD development cohort ($n = 15$).

excretion during the development of azotemic CKD (ie, 12 and 6 months before and at time of azotemic CKD diagnosis) was of specific interest and compared to the healthy cohort at each timepoint (ie, at timepoints –12 months, –6 months, and maintained health). To this end, we employed a targeted metabolomic approach aimed at analyzing (I) specific gut-derived uremic toxins, (II) their intermediates, and (III) dietary precursors. Such an approach allows for the elucidation of metabolic changes in uremic toxin-related pathways that are associated with development of overt CKD in cats.^{1,2} In this pathophysiological context, respectively (I) indoxyl-sulfate, TMAO, and p-cresyl-sulfate; (II) indole, trimethylamine (TMA), and p-cresol; and (III) tryptophan, carnitine, betaine, and tyrosine emerged as important metabolites.^{19–21} Therefore, they were selected to assess their evolution in preazotemic CKD stages.

2.3 | Sample collection

Blood samples were acquired by venipuncture, specifically from the cephalic or jugular veins (23-gauge needle), and collected into plain serum tubes (VACUETTE Serum Tubes with Gel). The samples were allowed to clot for maximum 15 minutes at room temperature, followed by centrifugation (5 minutes; 1931×g) at 4°C. Within a narrow time window (15 minutes), sterile urine samples were obtained by ultrasound-guided cystocentesis, utilizing a 22-gauge needle and transferred to plain tubes. The samples then were centrifuged (3 minutes; 355×g) at room temperature. After centrifugation, both serum and urine supernatants were aliquoted into Eppendorf tubes and preserved at –80°C until further analysis.

2.4 | Metabolomics analyses

All 10 analytical standards (Table S1) and internal standard (ISTD) were purchased from reputable sources (Sigma-Aldrich [St Louis, Missouri, USA], ICN Biomedicals Inc [Ohio, USA]).

All solvents utilized in this experiment (ie, acetonitrile, acetone, and methanol) were of liquid chromatography-mass spectrometry grade for extraction purposes and sourced from Thermo Fisher Scientific (Loughborough, UK) and VWR International (Merck, Darmstadt, Germany). Ultrapure water was obtained by the utilization of a purified water system, Arium 611UV (Sartorius, Göttingen, Germany).

We designed, optimized, and validated specific metabolomics extraction protocols for feline serum and urine samples targeting the selected metabolites.³⁰ For both biomatrices (urine, serum), the same ultrahigh-performance liquid chromatography (UHPLC) and high-resolution mass spectrometry (HRMS) settings were applied.^{1,30} For chromatographic separation, a Vanquish Horizon UHPLC system (Thermo Fisher Scientific, San José, CA, USA) was used with an Acquity UHPLC HSS C18 column T3 1.8 µm (150 mm × 2.1 mm; Waters, Manchester, UK) at a column temperature of 45°C. An injection volume of 10 µL at a maximum injection time of 150 ms was applied. Ultrapure water and acetonitrile, both acidified with 0.1%

formic acid, were used as analytical solvents. A constant flow rate of 0.4 mL/min was maintained throughout the analysis. An Exploris 120 stand-alone bench top orbitrap HRMS (Thermo Fisher Scientific, San José, CA, USA), equipped with a heated electrospray ionization source (HESI II), operating in polarity switching full scan mode at a scan range of 53–800 Da and with a mass resolution of 120 000 full width at half maximum (1 Hz), was used. Before mass spectrometric detection, initial instrument calibration was achieved by infusing ready-to-use calibration mixtures (Thermo Fisher Scientific, San José, CA, USA).

For each biomatrix, quality control samples (QCs) were generated by pooling sample extracts specific to the respective matrix. To adjust for instrumental fluctuations, QCs were run in duplicate after every 10 samples. Furthermore, at the start of the analyses, samples from the QC pool were injected to facilitate column conditioning, tailored to the specific biomatrix under investigation. An analytical standard mixture containing all a priori selected analytical standards (n = 10); (Table S1) was injected before, in the middle, and at the end of the analyses. This procedure enabled the assessment of the operational parameters of the device and ensured accurate identification of the designated metabolites.

2.5 | Data analysis

Targeted data processing was carried out using Xcalibur 3.0 (Thermo Fisher Scientific, San José, CA, USA). In this process, the identification of compounds relied on their *m/z*-value, carbon-isotope (C-isotope) profile, MS/MS fragmentation spectra, and retention time in relation to that of the internal standard.^{30,31} Compound abundance was determined based on the total area under the curve. Subsequently, the data underwent QC normalization. In the case of urine data, an additional normalization step was applied, where the data was adjusted by urinary creatinine concentration.^{32,33} We employed R (version 4.2.1) and SPSS Statistics 28 for statistical analyses. Group comparisons were conducted using QC-normalized abundance, without absolute quantification of metabolites.

Serum and urine abundance of gut-derived uremic toxins and their precursors were compared between the healthy cohort and the CKD development cohort at 3 timepoints: 12 months prior, 6 months prior, and at the time of azotemic CKD onset or maintained health. To do so, a nonparametric independent Mann-Whitney *U*-test (for non-normally distributed data) or students *t*-test was employed (for normally distributed data). The Friedman test was applied to evaluate the progression of specific metabolites over time in the CKD development cohort, using only data from those cats with available measurements at all timepoints (ie, 12 months prior, 6 months prior, and at the onset of azotemia). Normality was evaluated using the Shapiro-Wilk test, and the data distribution was visualized using histograms and QQ-plots. To address the issue of multiple comparisons, a post hoc correction procedure based on false discovery rates (FDR), specifically the Benjamini-Hochberg method, was applied. To assess the predictive ability and associations of circulating uremic toxins (indoxyl-sulfate,

p-cresyl-sulfate, and TMAO) with the development of azotemic CKD, univariate binary logistic regression analysis was conducted. Each model was constructed to forecast the binary outcome for CKD development vs remaining healthy (control group) based on the abundance of the specific uremic toxins: indoxyl-sulfate, TMAO, and p-cresyl-sulfate, assessed 6 months before the onset of azotemic CKD. Age, sCr, SDMA, USG, and UPC were included in our logistic regression analysis because they were reasoned to confound the association between the selected uremic toxins and CKD. Especially, the influence of decreasing glomerular filtration rate (GFR), estimated by sCr and SDMA and the effect of renal tubular function, estimated by USG and UPC on the circulating abundance of the selected gut-derived uremic toxins were of specific interest before development of azotemic CKD. The effect of age per se was added to exclude potential effects other than renal function on circulating gut-derived uremic toxins. Spearman correlation analysis was performed to assess the underlying relationship between the studied gut-derived uremic toxins and the selected covariates in the prediction of azotemic CKD onset.

3 | RESULTS

3.1 | Inclusion

The eligible population included 171 cats that appeared healthy and initially were presented by their owners. Of these, 127 cats were included as the study population for longitudinal follow-up. The main reasons for exclusion at baseline were hyperthyroidism ($n = 25$), FIV-positive status ($n = 9$), preexisting CKD diagnosis ($n = 6$), and other morbidities ($n = 4$). After the first follow-up at 6 months, 7 cats developed azotemic CKD; after 12 months, an additional 10 cats; at 18 months, 1 cat; and after 24 months, another 4 cats. Consequently, a total of 22 cats developed CKD, with backtracked samples and data available for 6 months before diagnosis, and for 15 cats, data was available 12 months before diagnosis. After 12 months, 72 cats from the initial cohort of healthy cats remained in the study. The main reasons for exclusion from the healthy group were CKD diagnosis ($n = 17$), development of other morbidities ($n = 13$), owner noncompliance ($n = 11$), death or euthanasia ($n = 8$), and hyperthyroidism ($n = 6$). Of these 72 eligible cats, 26 cats were randomly selected and backtracked as the control cohort, meeting the specified criteria regarding renal function (Figure 1).

3.2 | Routine renal parameters

The CKD development cohort was on average 2 years older than the healthy control cohort at baseline (Table 1). Body weight did not differ between groups at baseline, 6 months before and at the time of CKD diagnosis, and no clear trend was observed regarding weight loss in the CKD development cohort before onset of azotemia (Table S2). Furthermore, no significant differences were found between groups regarding SBP at all timepoints. Symmetric dimethylarginine but not

sCr was significantly higher at baseline in the CKD development cohort. However, 6 months before azotemia developed, both SDMA and sCr were higher in the CKD group compared with the healthy cohort. Importantly, even at this timepoint, SDMA remained within the reference range ($8\text{--}14\text{ }\mu\text{g/dL}$) for 82% (18/22) of the cats (Table 1). Serum creatinine concentrations showed a clear continuous increase from baseline to the development of azotemia (Table 1), particularly during the 6 months leading up to the onset of azotemia (Table S2). The SDMA concentrations began to increase only between 6 months prior and at the time of CKD diagnosis (Table S2), without showing a significant overall increasing trend during the 1-year period (Table 1). Urinary specific gravity was markedly decreased at baseline in the CKD development cohort, falling around the threshold of optimal urine concentration, and showing a significant downward trend over time within this group (Table 1). Conversely, UPC showed no differences between the 2 groups at any timepoint and remained stable over time within the CKD development cohort (Tables 1 and S2).

3.3 | Uremic toxins and associated metabolites

In the indole pathway, both the bacterial tryptophan catabolite indole and its uremic toxin derivative, indoxyl-sulfate, showed significantly increased serum, but not urine, abundance 6 months before CKD diagnosis compared with the healthy controls (Tables 2, 3, and Figure 2). These increases in serum abundance were even more pronounced at the time of diagnosis. Over the different timepoints, a significantly increasing pattern in serum indole abundance was observed within the CKD development cohort, especially between 12 and 6 months before diagnosis (Tables 2 and S3). Despite indoxyl-sulfate having a relatively high abundance already observed at baseline, a significant increase was observed between 12 months prior and the time of CKD diagnosis (Table S3). However, changes in indole and indoxyl-sulfate abundance 12 months before the onset of azotemia remained statistically insignificant in comparison with the healthy cohort. Tryptophan showed a significant decrease in urinary abundance at the time of azotemic CKD diagnosis, but no significant changes in serum were observed.

A significant increase in serum TMAO, but not TMA was observed 6 months before the onset of azotemia (Table 2 and Figure 2). Trimethylamine-N-oxide also showed augmented urinary excretion at that time (Table 3). At the time of diagnosis, the serum abundance and urinary excretion of TMAO further increased and serum trimethylamine (TMA) also became significantly increased. Both serum and urine TMAO showed significant upward patterns during azotemic CKD development, especially between 12 and 6 months before diagnosis for serum abundance (Tables 2, 3, S3, and S4). Serum carnitine and betaine showed no clear changes in abundance between the CKD and control group, but, a significant upward trend was observed for carnitine over time within the CKD development cohort attributed to the increase between 6 months prior and the time of diagnosis (Tables 2 and S3). Also, serum betaine abundance decreased in the CKD cohort over time, especially between 6 months prior and the time of diagnosis (Table S3).

TABLE 1 Comparison of baseline routine parameters related to renal health between a CKD development cohort and a cohort of healthy controls.

	Healthy cohort		CKD development cohort		P value (between groups) ^a
	Mean (SD)	Median (IQR)	Mean (SD)	Median (IQR)	
Age (years)					
–12 months	11.7 (1.3), n = 26	11.5 (11–13)	13.3 (2.8), n = 15	13.5 (12–15)	.08 ^b
Body weight (kg)					
–12 months	4.79 (1.14), n = 26	4.50 (3.95–5.16)	4.33 (1.03), n = 15	3.97 (3.55–4.95)	.14 ^c
–6 months	4.71 (1.11), n = 26	4.55 (3.89–5.10)	4.45 (0.96), n = 22	4.27 (3.80–4.80)	.71 ^c
–Healthy/CKD	4.61 (1.08), n = 26	4.55 (3.79–5.00)	4.30 (1.03), n = 22	4.12 (3.45–4.88)	.71 ^c
P value (within CKD group n = 15) ^d				.247	
Systolic blood pressure (mm Hg)					
–12 months	147 (26), n = 23	150 (125–160)	168 (19), n = 13	160 (150–185)	.06 ^b
–6 months	152 (25), n = 23	150 (130–170)	155 (19), n = 17	150 (143–165)	.68 ^b
Healthy/CKD	146 (20), n = 21	150 (126–160)	156 (22), n = 19	160 (140–170)	.17 ^b
P value (within CKD group n = 15) ^d				.231	
Creatinine (μmol/L)					
–12 months	119.6 (21.3), n = 26	121.6 (104.3–137.2)	131.9 (17.9), n = 15	130.8 (125.2–145.9)	.09 ^b
–6 months	115.1 (14.8), n = 26	112.8 (105.2–125.7)	138.6 (15), n = 22	137.9 (130–152.5)	<.01 ^b
Healthy/CKD	122.7 (16.4), n = 26	125.5 (114–131.3)	197.3 (39.2), n = 22	183.9 (174.6–199.4)	<.01 ^c
P value (within CKD group n = 15) ^d				<.001	
Symmetric dimethylarginine [SDMA] (μg/dL)					
–12 months	9.5 (2.3), n = 26	9.4 (7.4–11.3)	11.3 (2.4), n = 15	11.5 (9.2–13.1)	.04 ^b
–6 months	8.8 (2.2), n = 26	8.35 (7.0–10.9)	11.6 (2.9), n = 22	11.2 (9.6–13.6)	<.01 ^b
Healthy/CKD	8.0 (2.0), n = 26	7.3 (6.7–9.8)	15.5 (7.4), n = 22	13.4 (10.5–18.4)	<.01 ^c
P value (within CKD group n = 15) ^d				.165	
Urinary specific gravity [USG]					
–12 months	1.047 (1.006), n = 26	1.046 (1.042–1.052)	1.038 (1.011), n = 15	1.036 (1.032–1.045)	.01 ^b
–6 months	1.048 (1.007), n = 26	1.047 (1.043–1.051)	1.031 (1.009), n = 22	1.030 (1.025–1.037)	<.01 ^b
Healthy/CKD	1.047 (1.007), n = 26	1.045 (1.042–1.051)	1.020 (1.007), n = 22	1.018 (1.015–1.025)	<.01 ^c
P value (within CKD group n = 15) ^d				<.001	
Urinary protein : creatinine [UPC]					
–12 months	0.16 (0.07), n = 24	0.15 (0.12–0.20)	0.17 (0.06), n = 14	0.17 (0.12–0.19)	.72 ^c
–6 months	0.16 (0.05), n = 25	0.16 (0.12–0.19)	0.18 (0.08), n = 21	0.16 (0.11–0.24)	.76 ^c
Healthy/CKD	0.16 (0.05), n = 26	0.14 (0.13–0.18)	0.15 (0.07), n = 19	0.14 (0.09–0.19)	.74 ^c
P value (within CKD group n = 15) ^d				.320	

Note: Comparisons were made between the CKD development cohort and the healthy cohort at timepoints –12 months, –6 months, and azotemic CKD onset or maintained health, respectively. Furthermore, comparisons were made in the CKD development cohort between the different timepoints to assess the behavior of the metabolites over time.

^aRepresents P value for comparison of the Healthy and CKD development cohort at timepoints –12 months, –6 months, and CKD onset or maintained health respectively, after FDR correction.

^bP value for comparison of normally distributed data, presented as mean/SD.

^cP value for comparison of not normally distributed data, presented as median/interquartile range (IQR).

^dRepresents P value for comparison of different timepoints in the CKD development cohort.

Tyrosine-derived uremic toxins p-cresol and p-cresyl-sulfate only increased in serum at the time of azotemic CKD diagnosis, without a significant upward pattern during CKD development (Table 2, S3, and Figure 2). No significant changes in urinary abundance were observed (Tables 3 and S4). Serum and urine tyrosine

also underwent no significant differences before the onset of azotemic CKD.

Univariate binary logistic regression analyses (Table 4) identified significant Chi-square statistics (indoxyl-sulfate: $\chi^2 = 7.369$, $P = .01$; TMAO: $\chi^2 = 9.069$, $P < .01$; p-cresyl-sulfate: $\chi^2 = 4.281$, $P = .04$),

TABLE 2 Comparison of serum abundance of specific gut-derived uremic toxins, their dietary precursors, and intermediates between a CKD development cohort and a cohort of healthy controls.

	Healthy cohort		CKD development cohort		P value (between groups) ^a
	Mean (SD)	Median (IQR)	Mean (SD)	Median (IQR)	
L-Tryptophan					
–12 months	0.91 (0.19), n = 26	0.92 (0.79–1.03)	0.82 (0.17), n = 15	0.83 (0.72–0.96)	.15 ^b
–6 months	0.87 (0.17), n = 26	0.87 (0.72–.99)	0.79 (0.18), n = 22	0.78 (0.66–0.95)	.18 ^b
Healthy/CKD	0.89 (0.16), n = 26	0.86 (0.78–1.00)	0.76 (0.18), n = 22	0.81 (0.66–0.90)	.08 ^b
P value (within group n = 15) ^c				.420	
Indole					
–12 months	0.62 (0.50), n = 26	0.53 (0.32–0.70)	1.60 (1.98), n = 15	0.67 (0.42–2.09)	.1 ^d
–6 months	0.81 (1.06), n = 26	0.32 (0.23–1.14)	1.73 (1.30), n = 22	1.26 (0.84–2.78)	.02 ^d
Healthy/CKD	0.61 (0.38), n = 26	0.46 (0.29–1.03)	2.09 (1.60), n = 22	1.39 (1.05–3.38)	<.01 ^d
P value (within group n = 15) ^c				.027	
Indoxyl-sulfate					
–12 months	0.83 (0.70), n = 26	0.69 (0.42–1.07)	1.18 (0.74), n = 15	1.00 (0.63–1.62)	.08 ^d
–6 months	0.83 (0.46), n = 26	0.83 (0.50–1.07)	1.44 (1.06), n = 22	1.01 (0.72–2.26)	.02 ^b
Healthy/CKD	0.76 (0.44), n = 26	0.72 (0.51–1.02)	2.02 (1.19), n = 22	1.76 (1.19–2.61)	<.01 ^d
P value (within group n = 15) ^c				.334	
Carnitine					
–12 months	1.10 (0.68), n = 26	0.94 (0.55–1.50)	1.09 (0.61), n = 15	0.92 (0.58–1.45)	1 ^d
–6 months	1.09 (0.77), n = 26	0.96 (0.46–1.54)	1.08 (0.63), n = 22	1.00 (0.51–1.50)	1 ^d
Healthy/CKD	1.08 (0.57), n = 26	1.04 (0.63–1.39)	1.48 (0.90), n = 22	1.23 (0.64–2.19)	.26 ^d
P value (within group n = 15) ^c				.011	
Betaine					
–12 months	0.89 (0.51), n = 26	0.62 (0.51–1.16)	0.94 (0.63), n = 15	0.62 (0.50–1.32)	1 ^d
–6 months	0.99 (0.51), n = 26	0.75 (0.59–1.24)	0.91 (0.66), n = 22	0.71 (0.48–1.08)	1 ^d
Healthy/CKD	1.08 (0.63), n = 26	0.93 (0.57–1.47)	0.90 (0.72), n = 22	0.60 (0.41–1.15)	1 ^d
P value (within group n = 15) ^c				.057	
Trimethylamine (TMA)					
–12 months	1.19 (0.77), n = 26	1.20 (0.98–1.29)	1.22 (0.37), n = 15	1.13 (1.05–1.31)	1 ^d
–6 months	1.13 (0.21), n = 26	1.09 (0.99–1.34)	1.17 (0.26), n = 22	1.17 (1.00–1.26)	.76 ^b
Healthy/CKD	1.10 (0.30), n = 26	1.08 (0.88–1.22)	1.37 (0.34), n = 22	1.29 (1.17–1.54)	.02 ^d
P value (within group n = 15) ^c				.449	
Trimethylamine-N-oxide (TMAO)					
–12 months	1.49 (0.77), n = 26	1.32 (0.87–1.95)	1.43 (0.63), n = 15	1.43 (1.01–1.65)	.79 ^b
–6 months	1.60 (0.62), n = 26	1.31 (0.70–1.64)	1.82 (1.80), n = 22	1.80 (1.33–2.31)	.01 ^b
Healthy/CKD	1.20 (0.61), n = 26	1.06 (0.78–1.68)	2.02 (0.87), n = 22	1.83 (1.58–2.14)	<.01 ^d
P value (within group n = 15) ^c				.031	
Tyrosine					
–12 months	0.90 (0.21), n = 26	0.89 (0.77–1.07)	0.86 (0.15), n = 15	0.84 (0.73–0.97)	.69 ^b
–6 months	0.85 (0.20), n = 26	0.85 (0.70–0.93)	0.89 (0.22), n = 22	0.86 (0.71–1.03)	.89 ^d
Healthy/CKD	0.91 (0.21), n = 26	0.88 (0.77–1.08)	0.83 (0.18), n = 22	0.83 (0.70–0.97)	.62 ^b
P value (within group n = 15) ^c				.127	
p-Cresol					
–12 months	0.99 (1.32), n = 26	0.60 (0.26–1.07)	1.24 (1.23), n = 15	0.96 (0.14–1.85)	.91 ^d
–6 months	0.63 (0.55), n = 26	0.53 (0.15–1.03)	1.33 (1.74), n = 22	0.83 (0.14–1.50)	.34 ^d
Healthy/CKD	0.71 (0.64), n = 26	0.55 (0.16–1.24)	1.55 (1.10), n = 22	1.47 (0.62–2.45)	.01 ^d
P value (within group n = 15) ^c				.449	

(Continues)

TABLE 2 (Continued)

	Healthy cohort		CKD development cohort		P value (between groups) ^a
	Mean (SD)	Median (IQR)	Mean (SD)	Median (IQR)	
p-Cresyl-sulfate					
–12 months	0.98 (1.26), n = 26	0.62 (0.27–1.09)	1.22 (1.18), n = 15	0.95 (0.16–1.85)	.93 ^d
–6 months	0.63 (0.54), n = 26	0.54 (0.15–1.01)	1.30 (1.67), n = 22	0.84 (0.15–1.50)	.32 ^d
Healthy/CKD	0.71 (0.62), n = 26	0.56 (0.17–1.26)	1.52 (1.07), n = 22	1.43 (0.63–2.42)	.01 ^d
P value (within group n = 15) ^c				.449	

Note: Comparisons were made between the CKD development cohort and the healthy cohort at timepoints –12 months, –6 months, and azotemic CKD onset or maintained health respectively. Furthermore, comparisons were made in the CKD development cohort between the different timepoints to assess the behavior of the metabolites over time.

^aRepresents P value for comparison of Healthy and CKD development cohort at timepoints –12 months, –6 months, and CKD onset or maintained health respectively, after FDR correction.

^bP value for comparison of normally distributed data, presented as mean/SD.

^cRepresents P value for Friedman's comparison of different timepoints in the CKD development cohort.

^dP value for comparison of not normally distributed data, presented as median/interquartile range (IQR).

indicating that circulating gut-derived uremic toxins individually serve as predictors for CKD risk 6 months before diagnosis. Furthermore, the pseudo- R^2 values of the models suggest that variations in indoxyl-sulfate, TMAO, and p-cresyl-sulfate account for approximately 19%, 23%, and 11.4% of the variance, respectively, in the development of azotemic CKD in cats. The odds ratios (ORs) indicated that for each unit increase in the abundance of indoxyl-sulfate (95% CI: 1.2–9.0, $P = .04$), TMAO (95% CI: 1.4–11, $P = .03$), and p-cresyl-sulfate (95% CI: 0.9–4.1, $P = .1$), the odds of developing CKD increase by 3.2, 3.9, and 1.9 times, respectively. However, the outcome of p-cresyl-sulfate was not significant.

The predictive significance of indoxyl-sulfate and TMAO changed upon adjusting for specific covariates related to renal health (Table 5). After accounting for sCr and SDMA, the predictive value of both indoxyl-sulfate and TMAO lost significance. More specifically, adjustment for USG strongly attenuated the relationship between indoxyl-sulfate and azotemic CKD development, but only mildly for TMAO. Age and UPC only slightly decreased the association between indoxyl-sulfate, TMAO, and the onset of azotemia. However, the predictive value of indoxyl-sulfate narrowly lost its significance when adjusted for age.

To elucidate the confounding effects on predictive capacity, the relationship between indoxyl-sulfate, TMAO, and selected clinico-pathological variables related to renal health was assessed (Table 6). Serum creatinine and SDMA concentrations showed a positive linear relationship with both indoxyl-sulfate and TMAO. Moreover, a negative correlation was found between USG and both indoxyl-sulfate and TMAO.

4 | DISCUSSION

Our main objective was to examine metabolic alterations in gut-derived uremic toxins (ie, indoxyl-sulfate, p-cresyl-sulfate, and TMAO) and related pathways associated with the development of azotemic

CKD in cats. A nested case-control design was applied, employing targeted metabolomics.

We showed that the serum abundance of indole, indoxyl-sulfate, and TMAO begins to increase at least 6 months before diagnosis of CKD in cats. In contrast, no significant changes were observed in serum or urine abundance of p-cresyl-sulfate at the onset of azotemic CKD. Indoxyl-sulfate, p-cresyl-sulfate, and TMAO rank among the most studied uremic toxins originating from the dietary precursors tryptophan, tyrosine, and carnitine, betaine, and choline, respectively. These dietary precursors undergo partial bacterial degradation in the gut, resulting in byproducts such as indole, p-cresol, and TMAO.^{3,4,34} After passing the intestinal tight junction barrier and entering circulation, these intermediates undergo metabolism in the liver, resulting in the formation of the aforementioned toxins.^{34,35} Excretion of indoxyl-sulfate, p-cresyl-sulfate, and TMAO involves renal tubular transporters, but a certain fraction is eliminated by glomerular filtration.^{34–36} Systemic accumulation of these toxins has been shown to occur in azotemic CKD in cats, worsening in more advanced disease stages.^{1,2}

Considering the dynamics of the indoxyl-sulfate pathway, several mechanisms may theoretically underlie this observation. In the context of impaired renal function, decreased tubular transport capacity, decreasing GFR or both may be the main underlying factors. Our results suggest that urinary excretion corresponds with higher circulating concentrations during the development of CKD. Despite this, excretion of indoxyl-sulfate may be unable to match further increasing abundance in the bloodstream at the time of azotemic CKD diagnosis. Moreover, in previous work, we demonstrated that the urinary abundance of indoxyl-sulfate, p-cresyl-sulfate, and TMAO remains unchanged in more advanced CKD stages, whereas serum concentrations were significantly correlated with CKD International Renal Interest Society (IRIS) stage.¹ These observations may be consistent with findings in human patients in whom lower fractional excretion of indoxyl-sulfate occurred at more advanced CKD stages, indicative of a

TABLE 3 Comparison of urine abundance of specific gut-derived uremic toxins, their dietary precursors, and intermediates between a CKD development cohort and a cohort of healthy controls.

	Healthy cohort		CKD development cohort		P value (between groups) ^a
	Mean (SD)	Median (IQR)	Mean (SD)	Median (IQR)	
L-Tryptophan					
–12 months	1.01 (.67), n = 26	0.93 (0.73–1.16)	0.80 (0.42), n = 15	0.70 (0.46–1.25)	.25 ^b
–6 months	1.01 (.67), n = 26	0.78 (0.60–1.11)	0.76 (0.44), n = 22	0.75 (0.40–0.96)	.29 ^b
Healthy/CKD	1.86 (4.53), n = 26	0.83 (0.53–1.39)	0.52 (0.32), n = 22	0.44 (0.29–0.72)	<.01 ^b
P value (within group n = 15) ^c				.091	
Indole					
–12 months	0.86 (0.61), n = 26	0.80 (0.45–1.12)	1.21 (0.85), n = 15	1.12 (0.46–1.90)	.71 ^b
–6 months	1.05 (0.72), n = 26	0.76 (0.52–1.35)	1.30 (0.97), n = 22	1.10 (0.60–2.13)	.75 ^b
Healthy/CKD	1.23 (1.04), n = 26	0.84 (0.53–1.56)	1.56 (1.48), n = 22	1.05 (0.52–2.05)	.57 ^b
P value (within group n = 15) ^c				.807	
Indoxyl-sulfate					
–12 months	0.99 (0.77), n = 26	0.91 (0.48–1.28)	1.22 (0.55), n = 15	1.11 (0.82–1.77)	.09 ^b
–6 months	0.96 (0.53), n = 26	1.03 (0.52–1.32)	1.56 (0.95), n = 22	1.44 (0.78–2.08)	.06 ^b
Healthy/CKD	1.08 (0.69), n = 26	0.92 (0.57–1.62)	1.51 (0.70), n = 22	1.43 (0.93–2.15)	.06 ^b
P value (within group n = 15) ^c				.420	
Carnitine					
–12 months	0.68 (0.72), n = 26	0.44 (0.27–.78)	0.80 (0.95), n = 15	0.40 (0.18–0.91)	.78 ^b
–6 months	0.97 (0.92), n = 26	0.56 (0.36–1.53)	0.63 (0.54), n = 22	0.51 (0.22–0.81)	.39 ^b
Healthy/CKD	0.95 (0.75), n = 26	0.70 (0.42–1.28)	0.92 (1.20), n = 22	0.48 (0.32–0.70)	.39 ^b
P value (within group n = 15) ^c				.936	
Betaine					
–12 months	0.47 (0.32), n = 26	0.37 (0.28–0.55)	0.87 (1.11), n = 15	0.47 (0.36–0.81)	.13 ^b
–6 months	0.73 (0.78), n = 26	0.43 (0.39–0.70)	0.89 (1.04), n = 22	0.60 (0.42–0.80)	.13 ^b
Healthy/CKD	0.65 (0.83), n = 26	0.47 (0.37–0.55)	0.95 (0.94), n = 22	0.58 (0.48–1.13)	.03 ^b
P value (within group n = 15) ^c				.155	
Trimethylamine (TMA)					
–12 months	1.02 (0.45), n = 26	0.93 (0.72–1.27)	1.38 (1.72), n = 15	0.98 (0.63–1.47)	.9 ^b
–6 months	0.85 (0.36), n = 26	0.78 (0.51–1.11)	1.62 (2.08), n = 22	1.08 (0.80–1.67)	.1 ^b
Healthy/CKD	1.24 (1.27), n = 26	0.98 (0.54–1.38)	1.29 (0.56), n = 22	1.13 (0.92–1.54)	.17 ^b
P value (within group n = 15) ^c				.165	
Trimethylamine-N-oxide (TMAO)					
–12 months	0.95 (0.28), n = 26	0.93 (0.73–1.10)	0.97 (0.44), n = 15	1.00 (0.71–1.32)	.92 ^d
–6 months	0.90 (0.32), n = 26	0.93 (0.65–1.08)	1.32 (0.71), n = 22	1.13 (0.91–1.88)	.05 ^b
Healthy/CKD	1.02 (0.46), n = 26	0.96 (0.71–1.33)	1.77 (0.81), n = 22	1.50 (1.26–2.15)	<.01 ^b
P value (within group n = 15) ^c				.002	
Tyrosine					
–12 months	0.91 (0.40), n = 26	0.85 (0.73–1.06)	0.76 (0.40), n = 15	0.63 (0.53–0.96)	.24 ^b
–6 months	1.00 (0.58), n = 26	0.90 (0.62–1.11)	0.76 (0.38), n = 22	0.71 (0.50–0.90)	.25 ^b
Healthy/CKD	1.58 (3.60), n = 26	0.80 (0.53–1.21)	0.70 (0.43), n = 22	0.58 (0.38–0.88)	.18 ^b
P value (within group n = 15) ^c				.127	
p-Cresol					
–12 months	1.01 (.83), n = 26	0.82 (0.42–1.28)	1.07 (0.84), n = 15	1.03 (0.25–1.84)	.86 ^b
–6 months	0.72 (0.46), n = 26	0.77 (0.23–1.11)	1.13 (1.00), n = 22	0.87 (0.21–1.77)	.43 ^b
Healthy/CKD	1.08 (1.32), n = 26	0.68 (0.29–1.53)	1.13 (0.70), n = 22	1.26 (0.55–1.61)	.83 ^b
P value (within group n = 15) ^c				.819	

(Continues)

TABLE 3 (Continued)

	Healthy cohort		CKD development cohort		P value (between groups) ^a
	Mean (SD)	Median (IQR)	Mean (SD)	Median (IQR)	
p-Cresyl-sulfate					
–12 months	1.01 (0.83), n = 26	0.82 (0.42–1.28)	1.07 (0.85), n = 15	1.02 (0.20–1.84)	.86 ^b
–6 months	0.72 (0.46), n = 26	0.76 (0.22–1.10)	1.13 (1.01), n = 22	0.88 (0.19–1.80)	.43 ^b
Healthy/CKD	1.08 (1.32), n = 26	0.69 (0.29–1.55)	1.13 (0.70), n = 22	1.25 (0.55–1.61)	.8 ^b
P value (within group n = 15) ^c				.819	

Note: Comparisons were made between the CKD development cohort and the healthy cohort at timepoints –12 months, –6 months, and azotemic CKD onset or maintained health respectively. Furthermore, comparisons were made in the CKD development cohort between the different timepoints to assess the behavior of the metabolites over time.

^aRepresents P value for comparison of Healthy and CKD development cohort at timepoints –12 months, –6 months, and CKD onset or maintained health respectively, after FDR correction.

^bP value for comparison of not normally distributed data, presented as median/interquartile range (IQR).

^cRepresents P value for Friedman's comparison of different timepoints in the CKD development cohort.

^dP value for comparison of normally distributed data, presented as mean/SD.

disproportionate decrease in tubular clearance of indoxyl-sulfate compared with GFR.^{35,37} These authors, therefore, suggested that tubular secretory mechanisms become saturated at higher CKD stages.³⁵ It can be hypothesized that decreasing GFR causes an initial retention of indoxyl-sulfate in the bloodstream, which is counteracted by increased tubular secretion, initially leading to increased urinary concentrations of indoxyl-sulfate. However, when CKD progresses compensatory tubular secretion becomes saturated and may further decrease, leading to worsening retention of indoxyl-sulfate in the circulation.^{35,38,39}

Other mechanisms also may be responsible for the early increase in indoxyl-sulfate. Interestingly, together with indoxyl-sulfate, its precursor indole also increases significantly during CKD development. This observation suggests that augmented bacterial formation or increased gut permeability or both may be considerable underlying factors along with decreased renal excretion.^{5,6,9} During azotemic stages of CKD, increasing urea concentrations lead to alterations in intestinal microbiota and disruption of the intestinal epithelial barrier.^{9,40} As a consequence, intestinal dysbiosis and shifts toward proteolytic indole- and p-cresol-forming microbiota are commonly observed in human patients with impaired renal function.^{39,40} More recently, decreased fecal microbiome diversity and richness also was confirmed in cats with CKD,⁵ whereby fecal isovaleric acid, a surrogate marker of excessive intestinal protein fermentation, was increased compared with healthy controls.⁴¹ However, to date intestinal dysbiosis has not been established in preazotemic CKD.

Furthermore, our results indicated that indoxyl-sulfate and TMAO were individually predictive for onset of azotemia 6 months before diagnosis. However, renal function biomarkers creatinine, SDMA, and USG were significantly correlated with indoxyl-sulfate and TMAO abundance, resulting in attenuated predictability after correction for these factors. Therefore, increasing indoxyl-sulfate and TMAO abundance likely reflects an already decreasing GFR rather than being causative factors initiating renal functional decline. On the contrary, in

2 studies of cats with established CKD, retention of indoxyl-sulfate was shown to be associated with impending CKD progression even after correction for renal function markers.^{8,42} At higher concentrations, indoxyl-sulfate starts to accumulate in tubular cells exerting direct toxicity via induction of cellular death and indirectly via promotion of oxidative stress and inflammation.^{43–45} Similarly, TMAO is associated with renal tubulointerstitial fibrosis in a continuous dose-dependent relationship.⁴⁶ As indoxyl-sulfate starts to increase early during the development of azotemia in cats, this toxin may be implicated in early pathophysiological mechanisms involved in the progression to overt CKD. In an animal model progressing from acute kidney injury to CKD, administration of indole exacerbated tubular inflammation and fibrosis, and the authors concluded that early indoxyl-sulfate inhibition is a useful strategy to prevent CKD development.⁴⁷ These results could not consistently be translated to human patients with CKD.⁴⁸ Nevertheless, in several human cohorts, decreases in circulating indoxyl-sulfate induced by PO sorbent treatment delayed deterioration of renal function and subsequent initiation of dialysis.⁴⁹

In human medicine, gut-derived uremic toxins also are recognized for their contribution to CKD-related morbidity, because they increase the risk of cardiovascular disease, contribute to renal anemia, negatively affect bone turnover, intensify muscle atrophy, and accelerate neuroinflammation-associated cognitive decline.^{50–52} Given their well-established toxic properties and association with negative disease outcomes, decrease of circulating indoxyl-sulfate and other gut-derived uremic toxins recently have received more attention in the therapeutic management of CKD in cats.¹⁰ In this context, different strategies have been investigated. Recent evidence suggests that dietary macronutrient composition directly impacts circulating concentrations of gut-derived uremic toxins in healthy adult cats.^{11,17,53} For example, cats fed high-protein diets had significantly increased circulating indoxyl-sulfate concentrations whereas tryptophan itself was not increased.¹⁷ Furthermore, because gut bacteria produce indole, a potential treatment strategy is the use of PO probiotics and

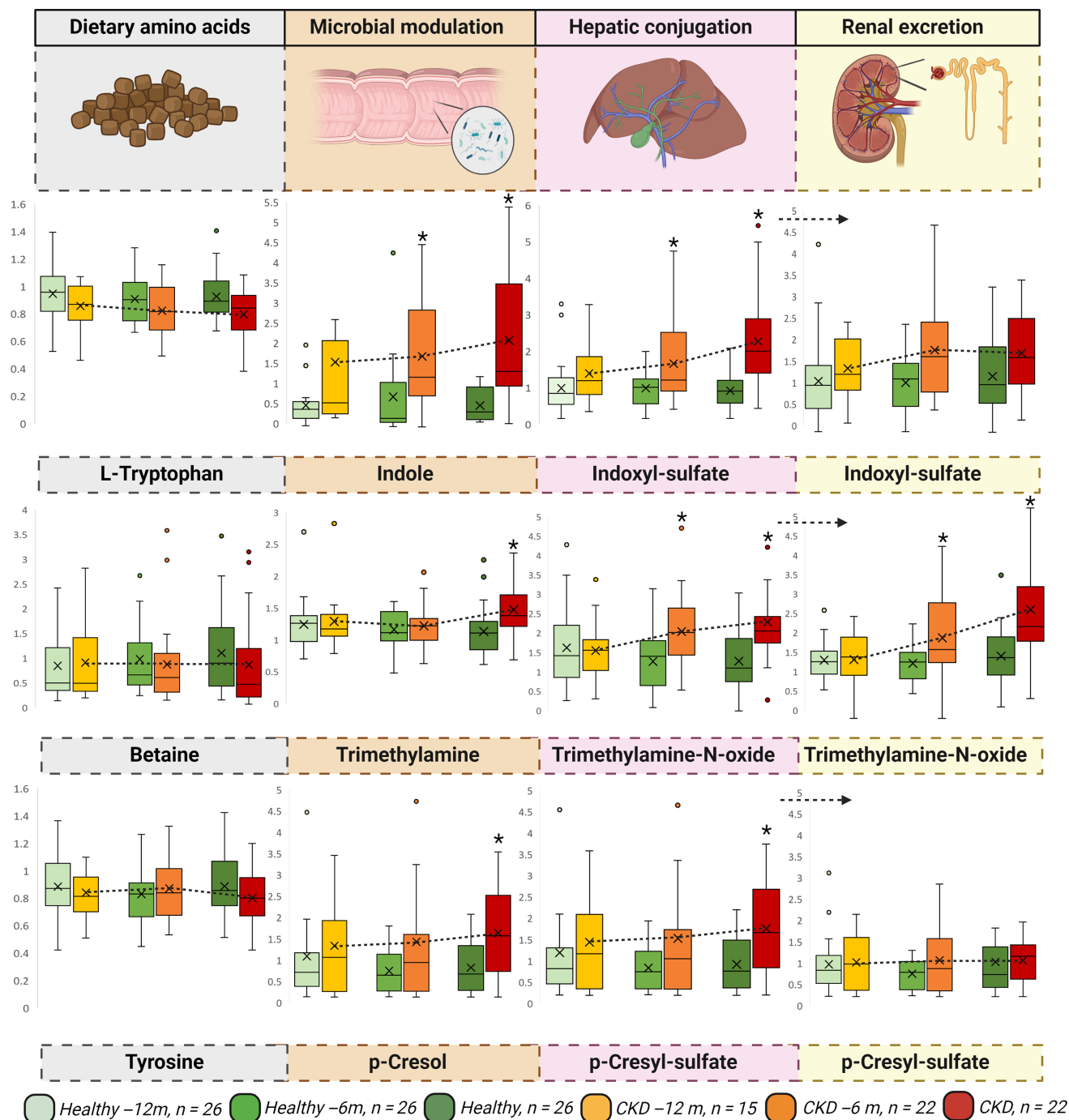


FIGURE 2 Comparisons of (I) serum dietary amino acids tryptophan, tyrosine, carnitine; (II) gut bacterial byproducts indole, p-cresol, trimethylamine (TMA); (III) related uremic toxins indoxyl-sulfate, p-cresyl-sulfate, trimethylamine-N-oxide (TMAO); and (IV) urinary abundance of these compounds between a healthy feline cohort and a cohort of cats developing azotemic CKD over time. Specific comparisons are shown between the healthy cohort and the CKD development cohort at 12 months, 6 months prior, and at the time of azotemic CKD diagnosis or maintained health. Significant P values ($<.05$) are represented by the symbol * after FDR correction.

prebiotics (eg, fermentable fiber) to alter the composition of the microbiome. However, so far, no consistent impact of pro- or prebiotics on indoxyl-sulfate concentrations in the bloodstream has been observed in cats in contrast to studies in humans.^{10,12,13} This difference may be a consequence of insufficient knowledge about potential modulation of the feline-specific microbiome. Especially

in cats with CKD, the microbiome appeared to be more resistant to fermentable fiber-induced change compared with healthy controls.¹³ Lastly, carbon-based adsorbents formulated to bind indole in the gastrointestinal tract showed promising potential in lowering circulating indoxyl-sulfate in cats consistent with findings in humans.^{14,54}

TABLE 4 Results of the binary logistic regression analysis exploring the associations between specific gut-derived uremic toxins and the onset of azotemic feline CKD 6 months before diagnosis.

	Model X^2 (P value)	Model's pseudo- R^2	Odds ratio	95% confidence interval	P value ^a
Indoxyl-sulfate	7.369 (.007)	0.190	3.2	1.2-9.0	.04
Trimethylamine-N-oxide (TMAO)	9.069 (.003)	0.230	3.9	1.4-11	.03
p-Cresyl-sulfate	4.281 (.039)	0.114	1.9	0.9-4.1	.1

^aRepresents P value after FDR correction.

Models	Indoxyl-sulfate			Trimethylamine-N-oxide (TMAO)		
	Odds ratio	95% CI	P	Odds ratio	95% CI	P
Model 1 ^a	3.003	0.992-9.009	.05	3.300	1.119-9.804	.03
Model 2 ^b	1.780	0.556-5.682	.33	1.538	0.461-5.128	.48
Model 3 ^c	2.250	0.791-6.410	.13	2.294	0.737-7.143	.15
Model 4 ^d	1.517	0.239-9.615	.66	6.544	0.847-50.000	.07
Model 5 ^e	2.920	1.050-8.130	.04	4.545	1.414-14.493	.01

^aModel 1 was adjusted for age.^bModel 2 was adjusted for serum creatinine.^cModel 3 was adjusted for serum SDMA.^dModel 4 was adjusted for urine-specific gravity (USG).^eModel 5 was adjusted for urine protein : creatinine ratio (UPC).**TABLE 5** Multivariable logistic regression analysis of indoxyl-sulfate and trimethylamine-N-oxide as predictors of azotemic CKD 6 months before diagnosis, adjusted for various renal health-related factors.

Variables	Indoxyl-sulfate		Trimethylamine-N-oxide (TMAO)	
	Spearman coefficient	P	Spearman coefficient	P
Age	0.074, n = 48	.618	0.298, n = 48	.04
Creatinine	0.395, n = 48	.01	0.534, n = 48	<.01
SDMA	0.389, n = 48	.01	0.498, n = 48	<.01
USG	−0.363, n = 48	.01	−0.414, n = 48	<.01
UPC	0.288, n = 46	.05	0.195, n = 46	.19

TABLE 6 Correlations between indoxyl-sulfate, trimethylamine-N-oxide, and selected clinicopathological variables related to renal health.

Abbreviations: UPC, urine protein : creatinine ratio; USG, urine-specific gravity.

A notable factor regarding interpretation of our study's results is the observation that sCr and especially USG were already significantly altered in the CKD development cohort compared with healthy controls 6 months before the diagnosis of renal azotemia. Additionally, this data underscores the potential usefulness of USG as a clinically applicable tool for serial monitoring of renal function in nonazotemic cats. Such changes may indicate that a substantial decrease in GFR was already present at that time in the CKD development cohort, but was not yet resulting in renal azotemia.²⁹ Notably, SDMA, proposed as an earlier indicator of decreasing GFR compared with sCr, stayed within the reference interval in 18 of 22 cats and remained below the IRIS threshold for stage 2 renal disease (<18 µg/dL) in all cats 6 months before azotemic CKD diagnosis.^{55,56} This observation provides further evidence that the demonstrated alterations in gut-derived uremic toxins appear to occur early in the disease continuum.

The primary limitation of our study was its small sample size, particularly for cats monitored 12 months before the onset of CKD. Given

the occurrence of many nearly significant P values, limited sample size may have hindered the achievement of statistical significance, especially when adjustments were made for multiple comparisons. Another limiting factor is that urine creatinine normalization can be dependent on factors (eg, age, muscle mass, hydration status) other than urinary dilution, and therefore these results should be interpreted with caution.^{32,33} Alternatives such as preanalytical dilution of urine specimens to a fixed creatinine concentration could have provided a more accurate representation of urinary compound abundance. Nonetheless, measuring compound abundance in a 24-hour urinary collection sample remains the gold standard, but is not feasible in a clinical setting.⁵⁷

In conclusion, we determined that the circulating gut-derived uremic toxins indoxyl-sulfate and TMAO increase at least 6 months before the diagnosis of azotemic CKD. However, a direct causal relationship between these toxins and the initiation of azotemic CKD in cats is still lacking. Nevertheless, collectively, our data supports that impaired gut-derived uremic toxin handling occurs early in the CKD continuum in cats, warranting early therapeutic intervention.

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CONFLICT OF INTEREST DECLARATION

Authors declare no conflict of interest.

OFF-LABEL ANTIMICROBIAL DECLARATION

Authors declare no off-label use of antimicrobials.

INSTITUTIONAL ANIMAL CARE AND USE COMMITTEE (IACUC) OR OTHER APPROVAL DECLARATION

Approved by the ethical committee of the faculties of Veterinary Medicine and Bioscience Engineering, Ghent University (EC2020-081).

HUMAN ETHICS APPROVAL DECLARATION

Authors declare human ethics approval was not needed for this study.

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